OpenMask3D:
Open-Vocabulary 3D Instance Segmentation

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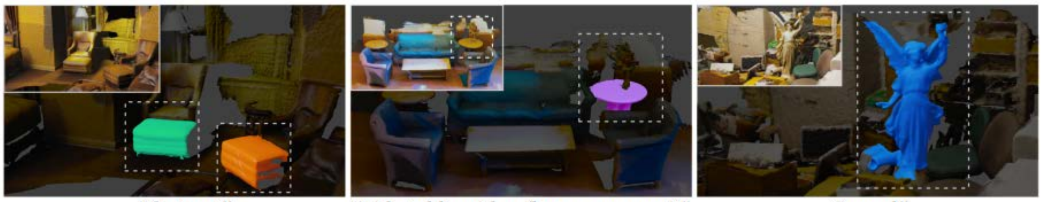
openmask3d.github.io



"footrest" *"side table with a flower vase on it"* *"angel"*

Figure 1: Open-Vocabulary 3D Instance Segmentation. Given a 3D scene *(top)* and free-form user queries *(bottom)*, our OpenMask3D segments object instances and scene parts described by the open-vocabulary queries.

Context

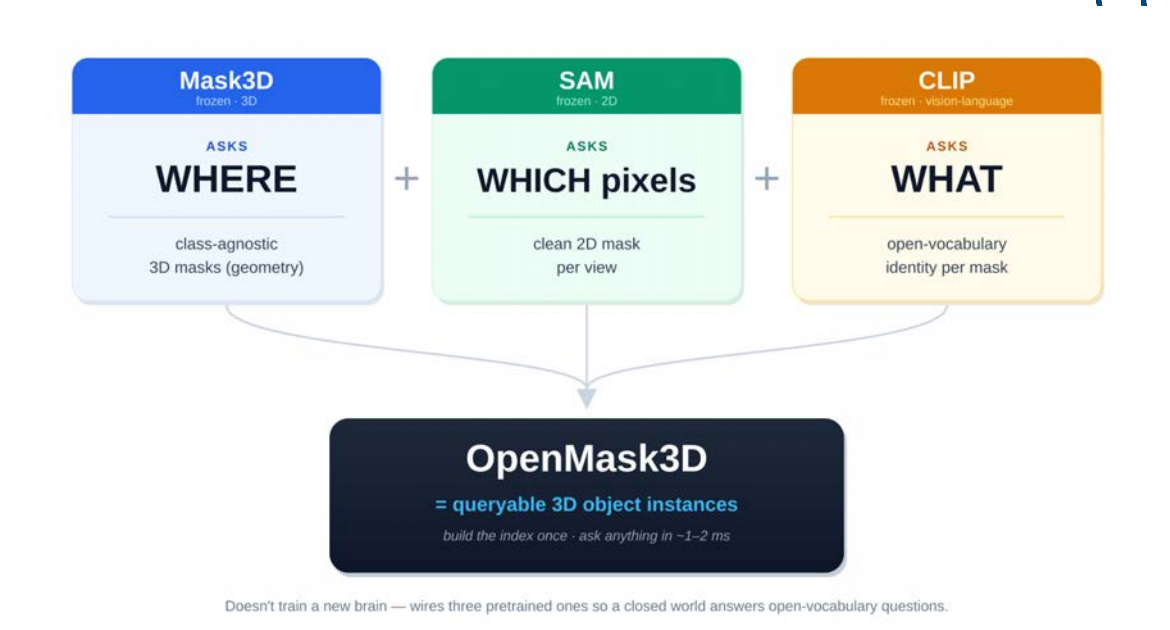
- Current approaches for 3D instance segmentation can typically only recognize object categories from a pre-defined closed set of classes that are annotated in the training datasets.
- limitations for real-world applications - objects that are not in the dataset.
- Open-vocabulary 3D instance segmentation.


"a table"

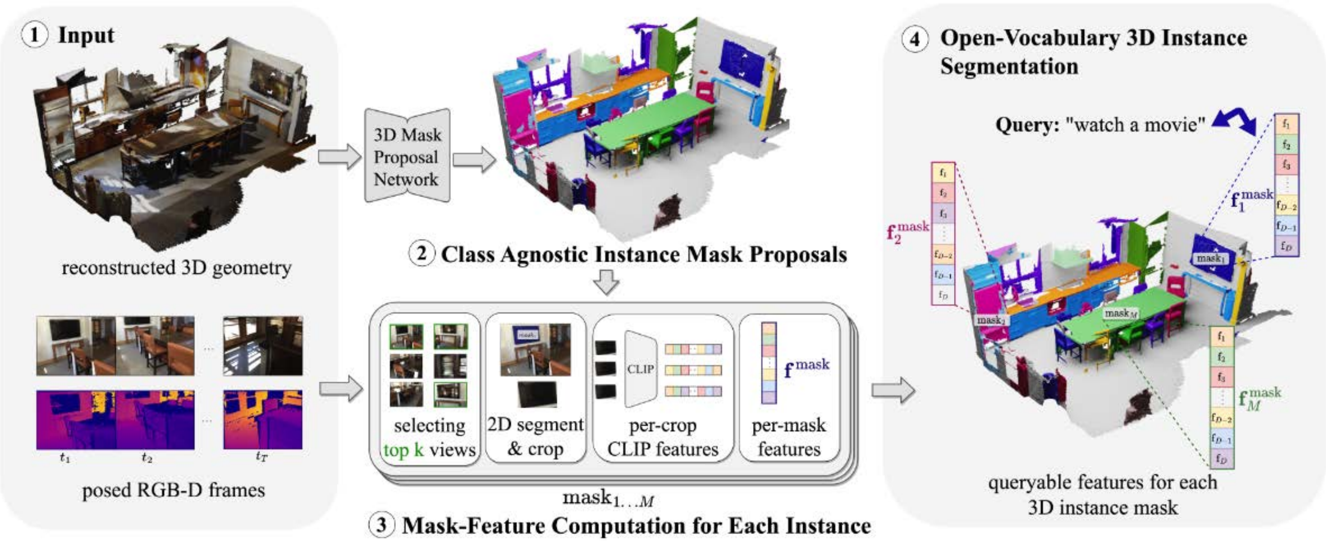

"a rectangular table"


“a side table with a flower vase on it”

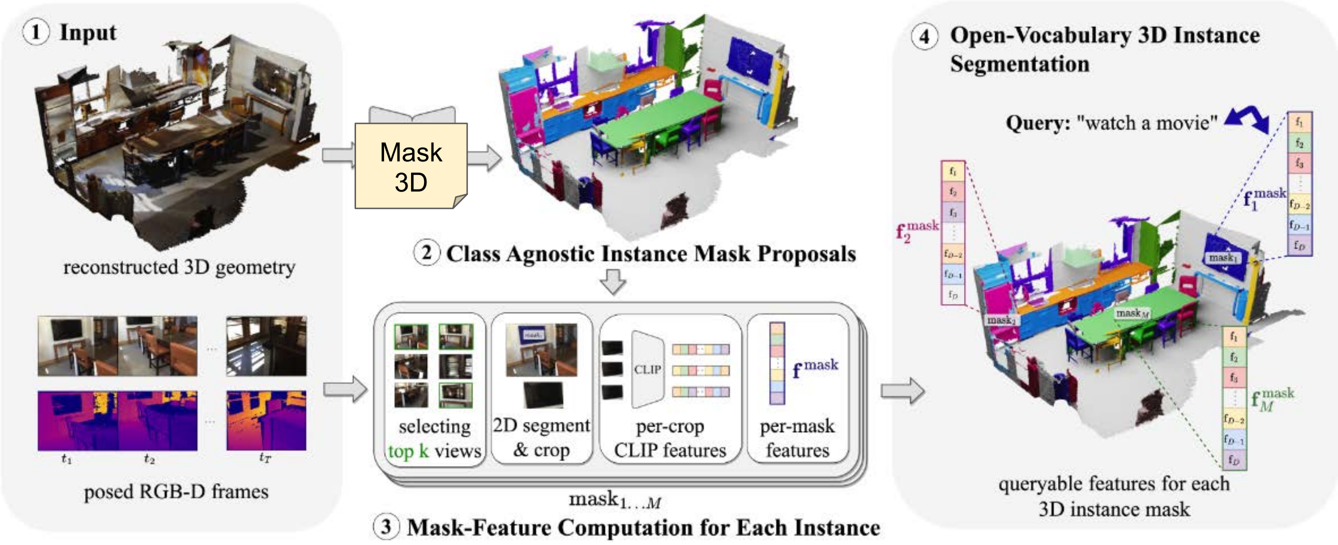

“a round, empty table”



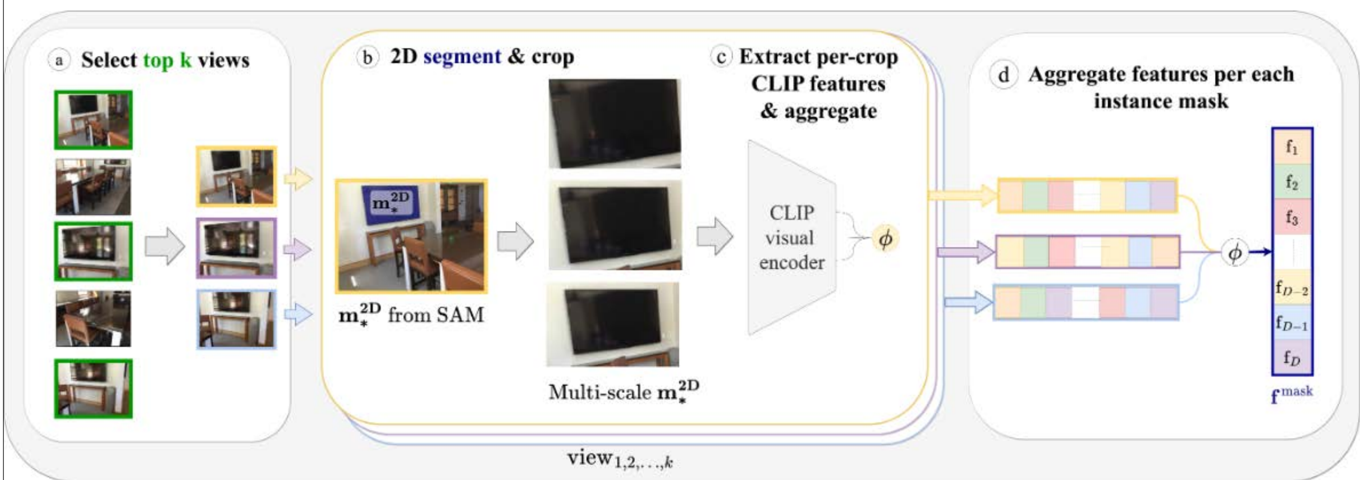
Architecture



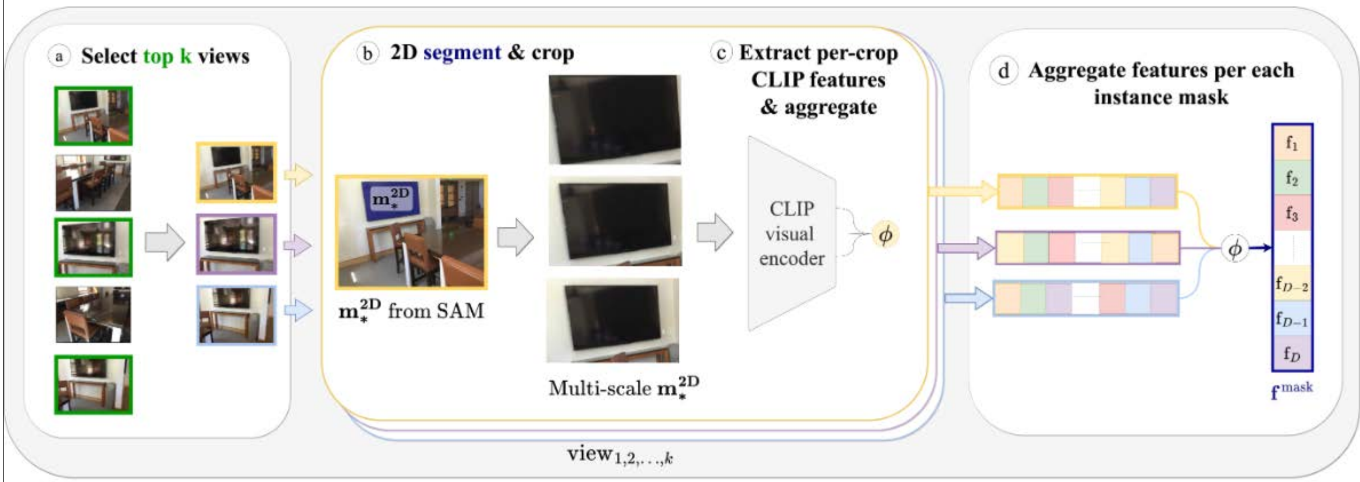
Architecture



Architecture

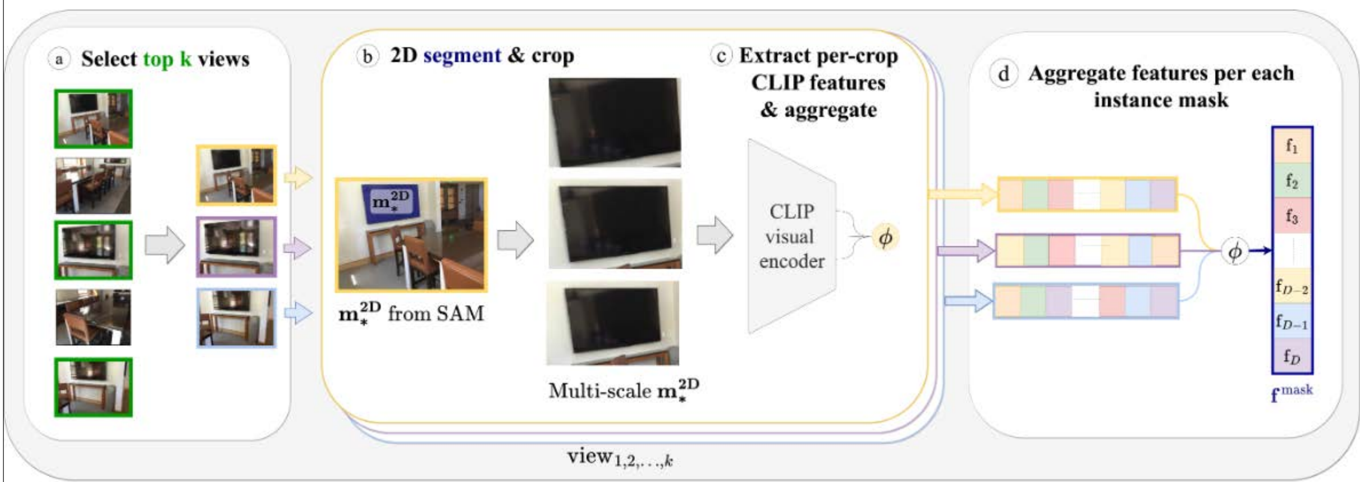


Architecture



(a) Selected by visibility score obtained using camera matrices

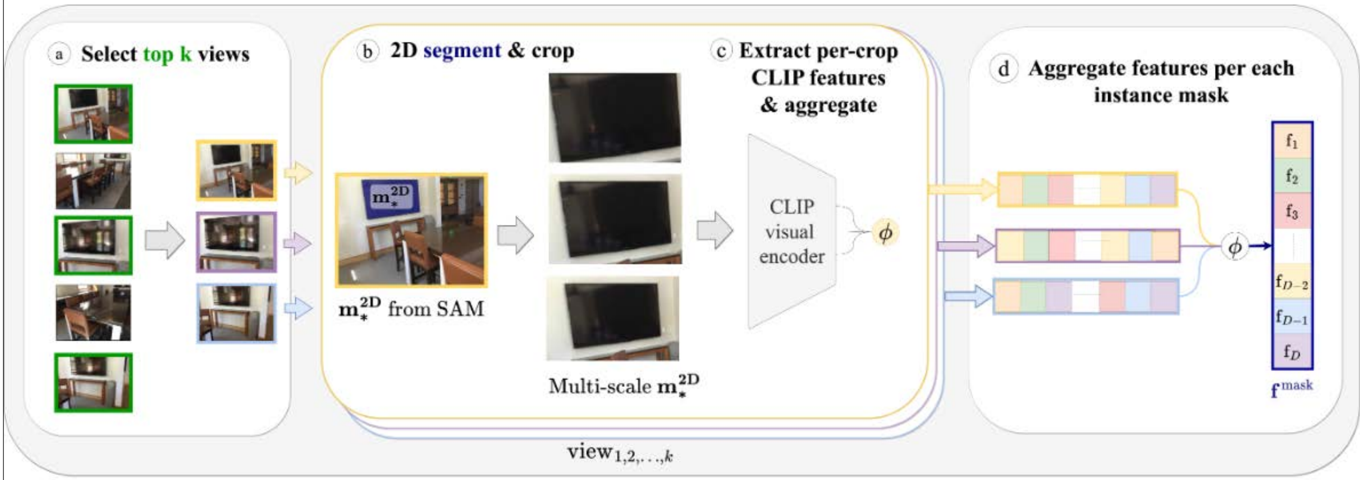
Architecture



(b) SAM generates r Masks and we select the best and do a L=3 multi-level crops

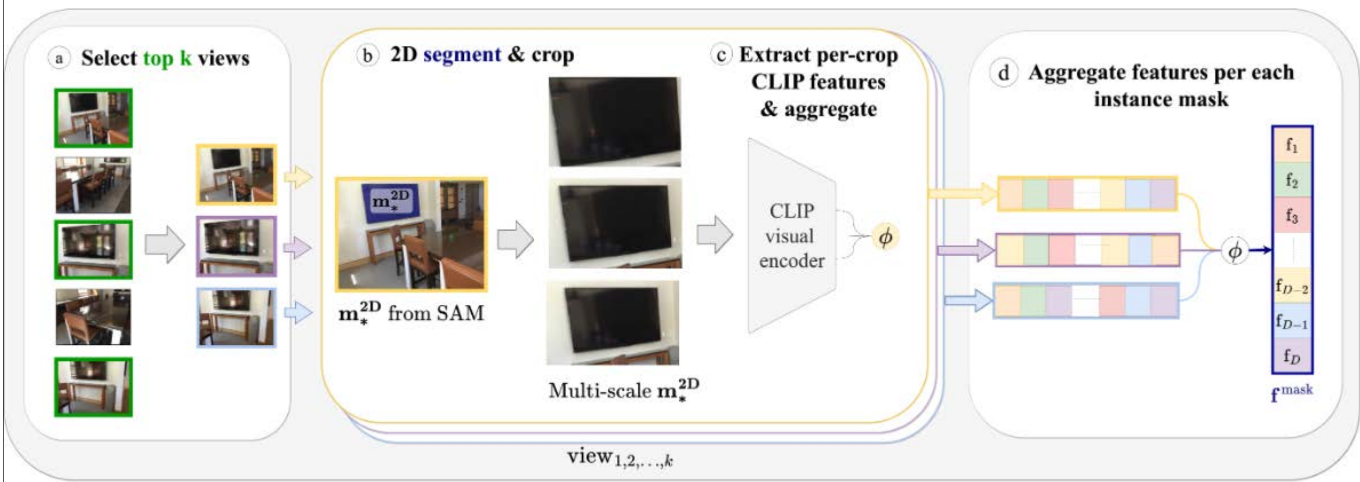
Centro Pi

Architecture



(c) Collected image crops are passed through the CLIP visual encoder

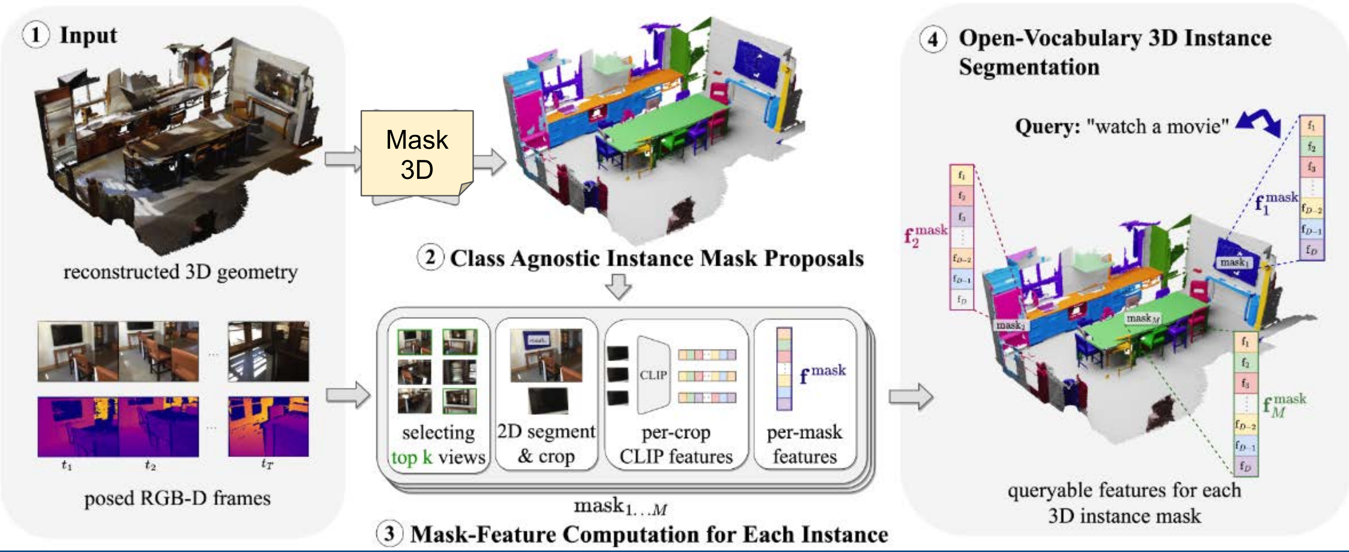
Architecture



(d) Average per-mask CLIP feature

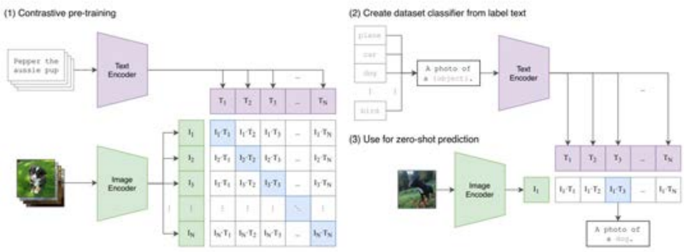
Centro Pi

Architecture

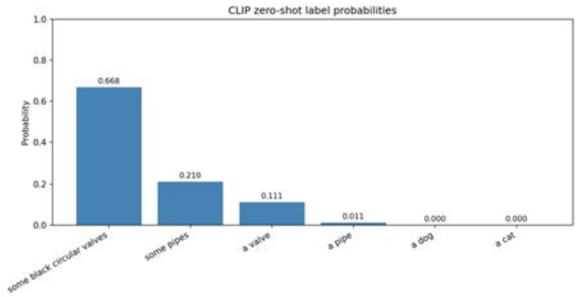


CLIP

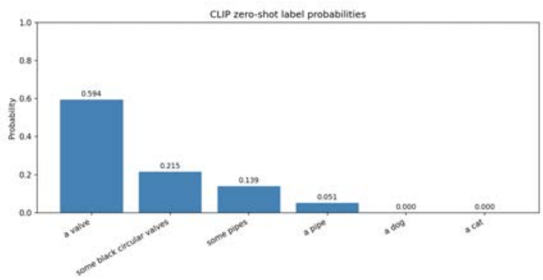
- Is the core engine that allows OpenMask3D to understand "open-vocabulary"
- Is uses the visual encoder together with the SAM to obtain feature vectors
- Create a single, task-agnostic mask-feature representation for each 3D instance



CLIP

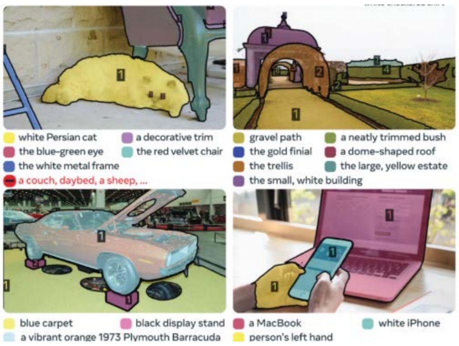

CLIP zero-shot label probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
0.668
0.210
0.111
0.011
0.000
0.000
some black circular valves
some pipes
a valve
a pipe
a dog
a cat

CLIP


CLIP zero-shot label probabilities
Probability
1.0
0.8
0.6
0.4
0.2
0.0
0.594
0.215
0.139
0.051
0.000
0.000
a valve
some black circular valves
some pipes
a pipe
a dog
a cat

CLIP

- Can be fine-tuned to our data
- SAM 3 already has prompt segmentation implemented
- It even returns bounding boxes



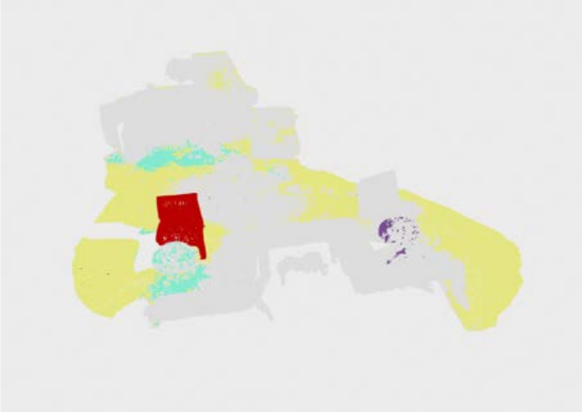
Scannet

3d reconstruction

Instances proposal

3d reconstruction

Query: “Chair”

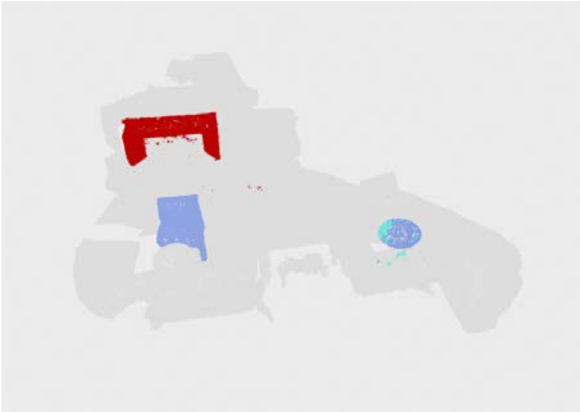


3d reconstruction

Query: “A place to sit”

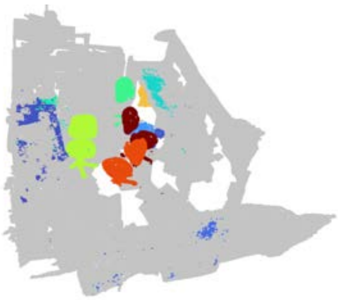
3d reconstruction

Query: “a place to put a cup of water”

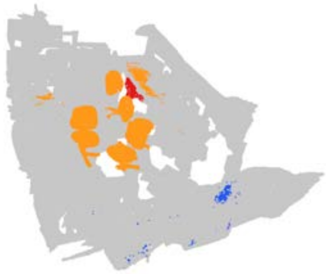


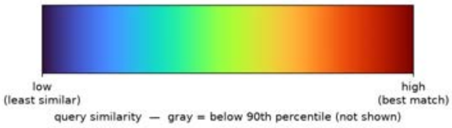
3d reconstruction

Query (local): “A chair”

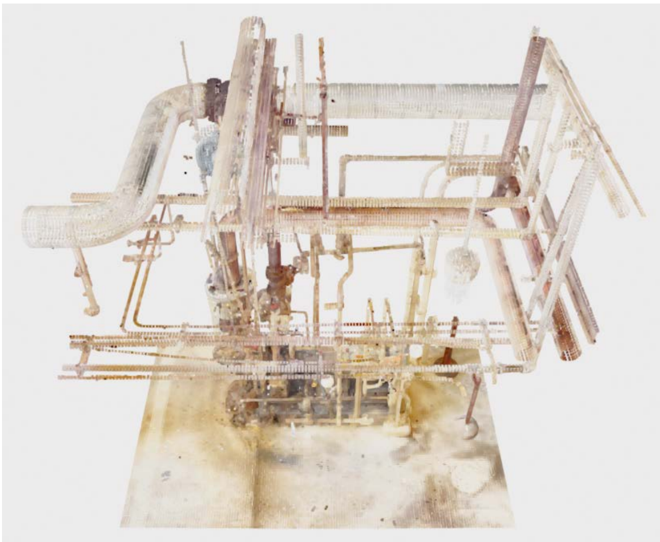


Query (global):
“Chair farthest from the door”

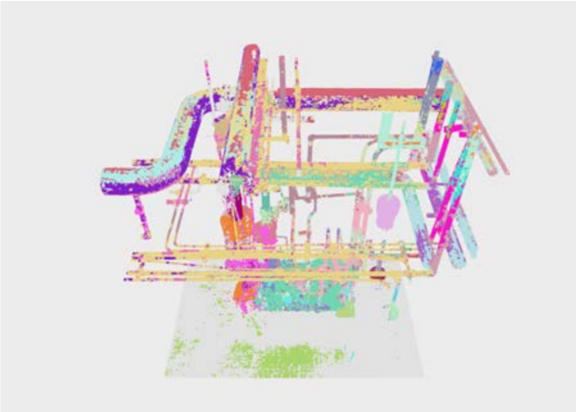


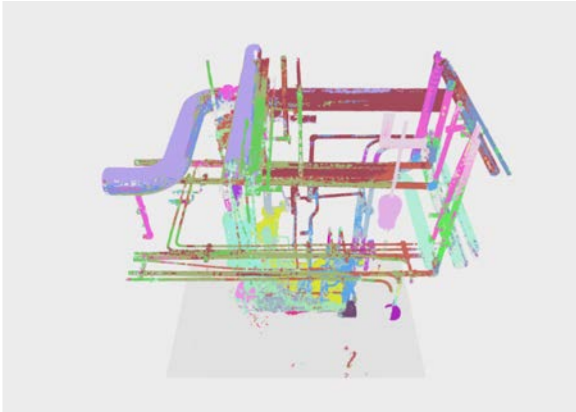


Radix dataset



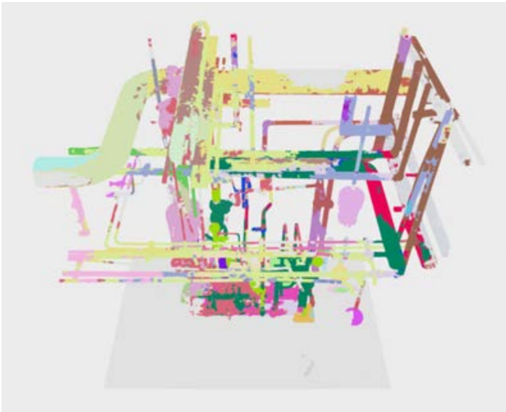
Initial segmentation examples:



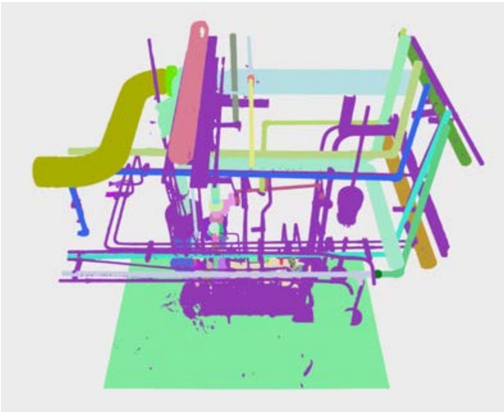


Centro Pi

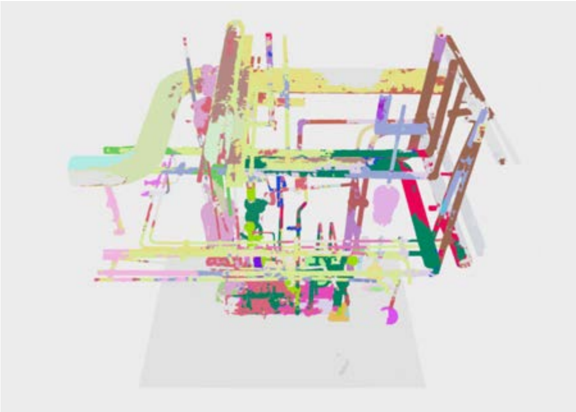
Baseline



Ground Truth



Centro Pi



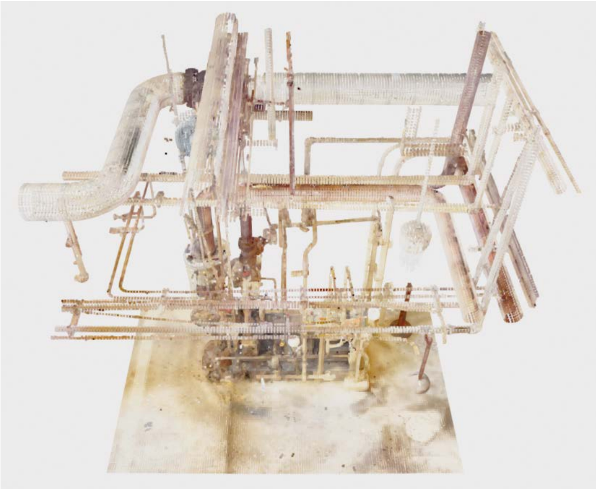
Voxel_size: Mask3d’s voxelization (adjusted to be resemble the training data).

Num_queries: number of instances the network may propose

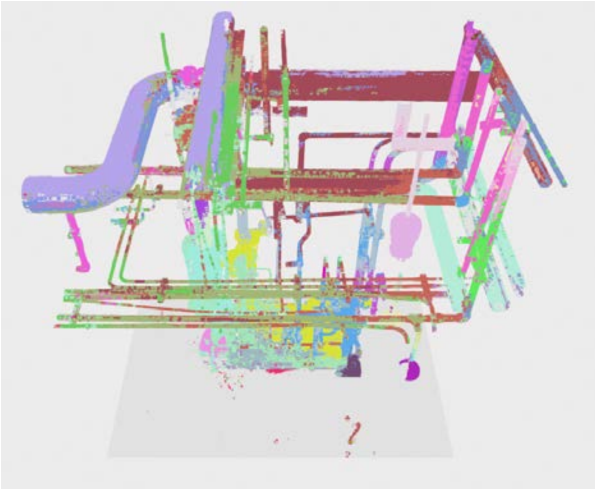
DBSCAN: post-processing that breaks instance proposals into connected components using point density.

`voxel`	`num_queries`	`dbscan_eps`	masks	coverage	fragmentation	note
20 mm	120	0.95	232	79 %	13.2	baseline (chosen in A)
20 mm	120	2.0	132	78 %	11.3	eps relaxes → pipes merge
20 mm	120	off	120	78 %	11.3	most coherent, deck still gray
30 mm	**120**	**off**	**120**	**87 %**	**13.1**	**chosen — coherent + deck filled**
30 mm	200	0.95	368	91 %	15.7	more queries → more fragments
20 mm	200	0.95	373	79 %	30.0	worst — over-segmented

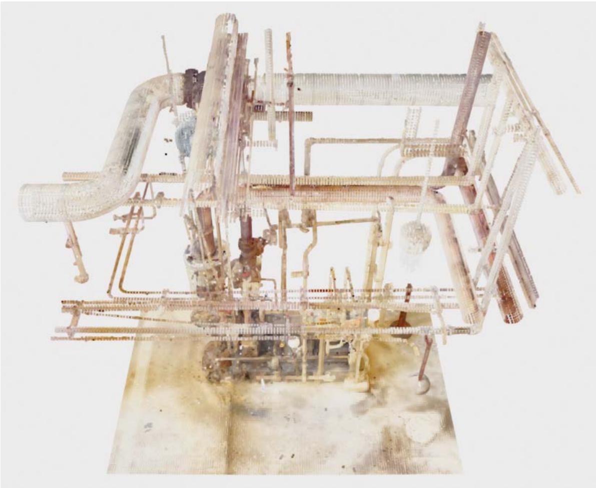
3d reconstruction



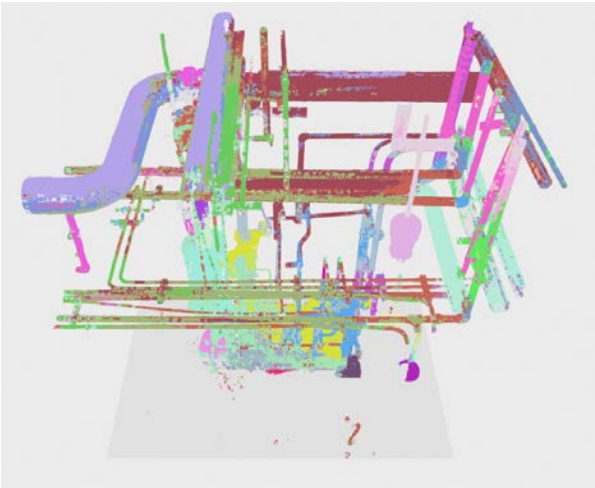
Instances proposal



3d reconstruction

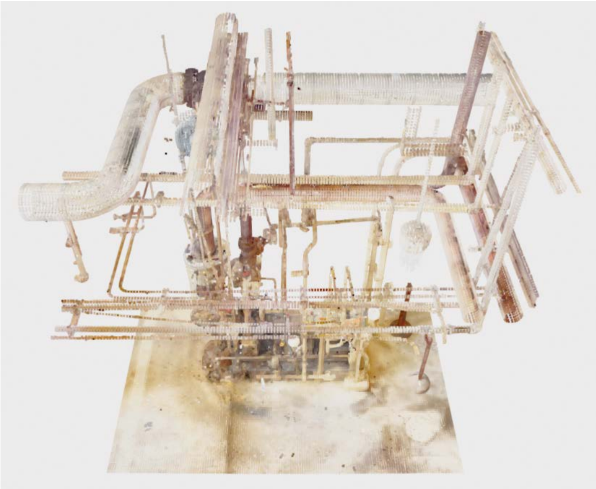


Instances proposal

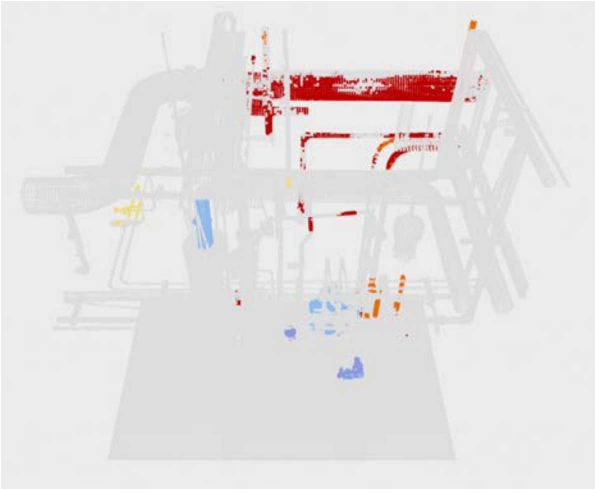


Centro Pi

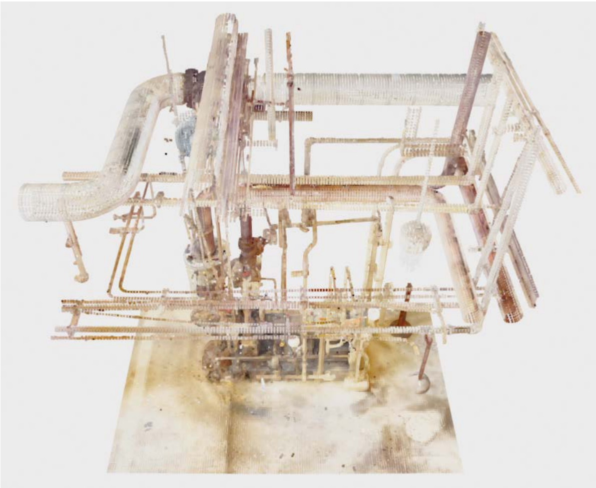
3d reconstruction



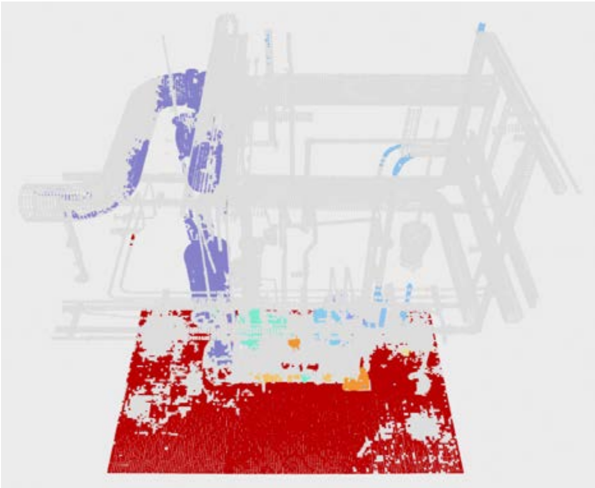
Query: “Pipe”



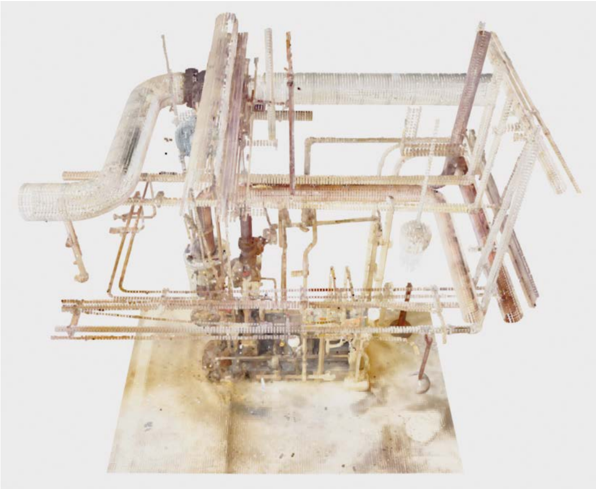
3d reconstruction



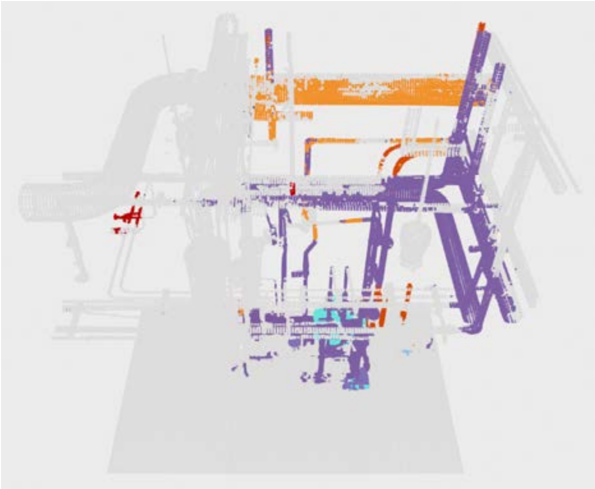
Query: “Environment”



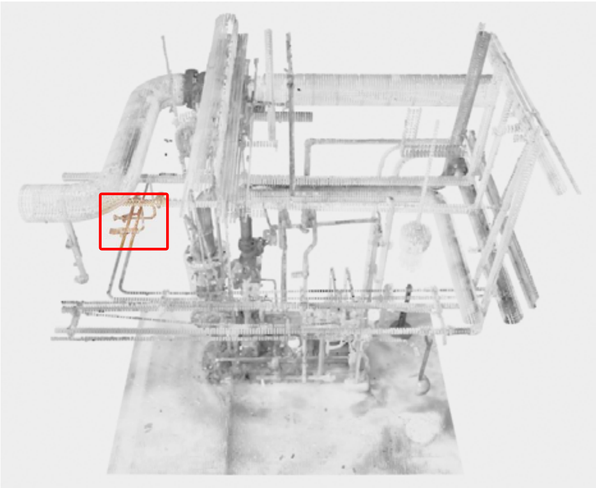
3d reconstruction



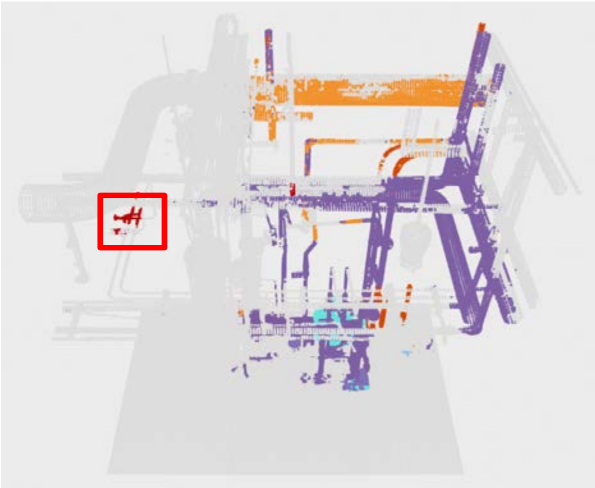
Query: “Valve”



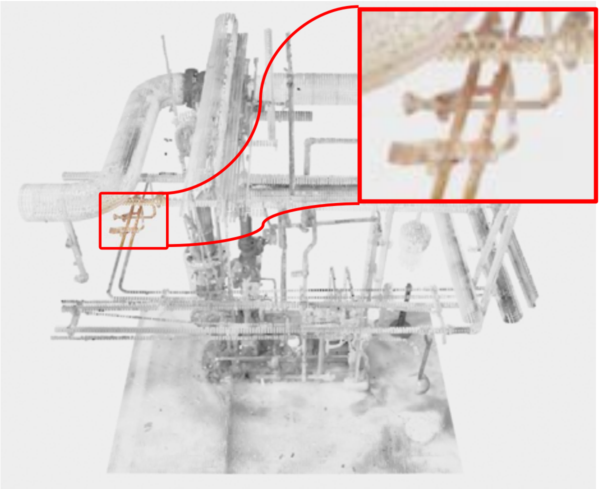
3d reconstruction



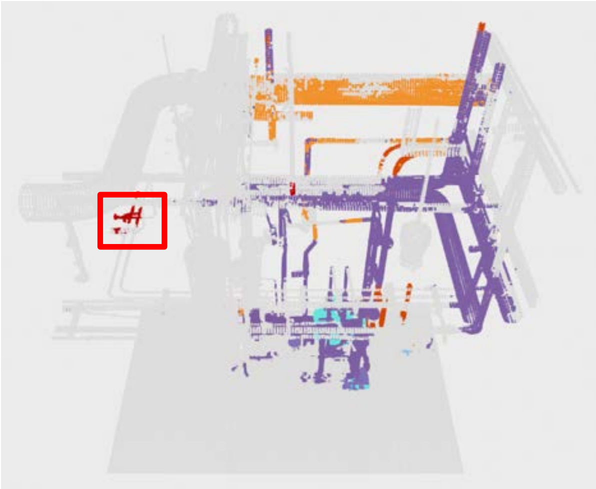
Query: “Valve”



3d reconstruction



Query: “Valve”



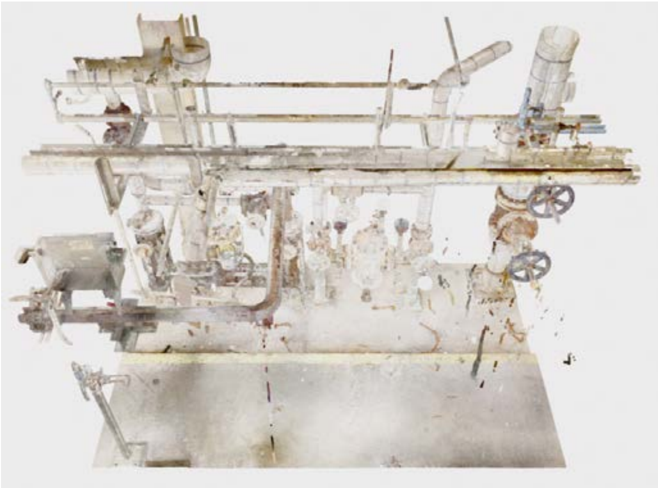
Next steps:

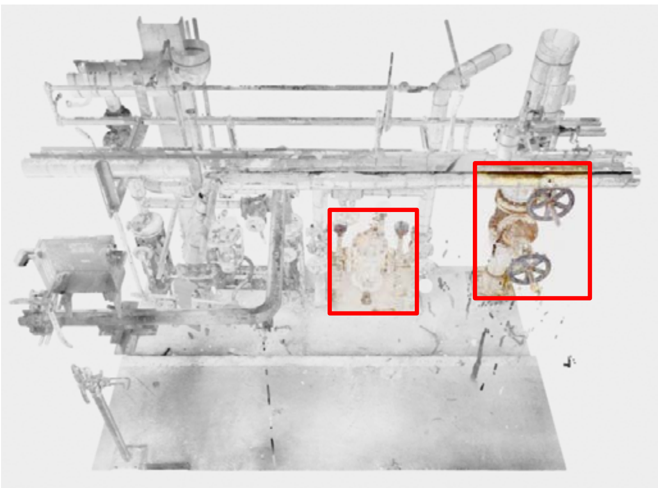
- Train Mask3d on Radix dataset
  Status: ready to launch.
  Plan: launch naive train and search for optimizable hyperparameters. Test different segmentations (decodes) available.

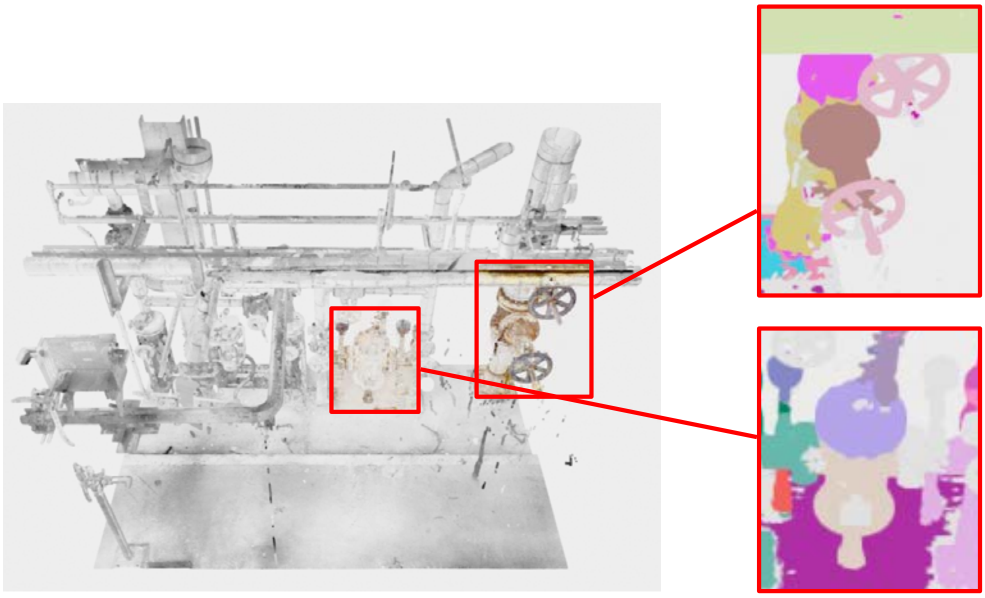
- Study ‘correct resolution’ and how to generate the views to use on Radix dataset for clip.
  Status: partially done; first study entangled Mask3d and Clip performances.
  Plan: test only CLIP. Use different resolutions, views and queries.

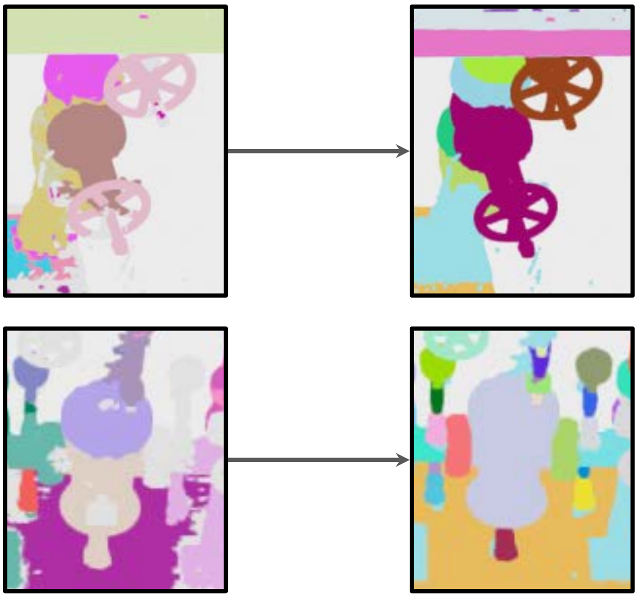
- Compare results with previous efforts (SoftGroup, SoftGroup+axis etc)
  Status: not started.
  Plan: —

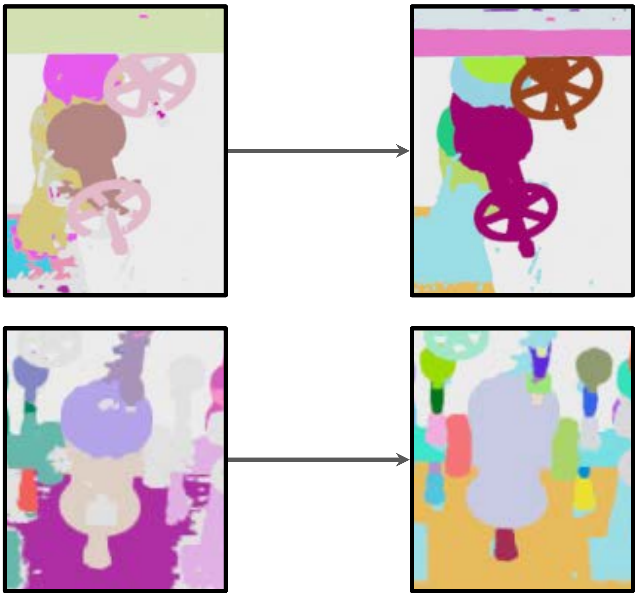
- Pursuit idea presented in next slides.
  Status: not started.
  Plan: —



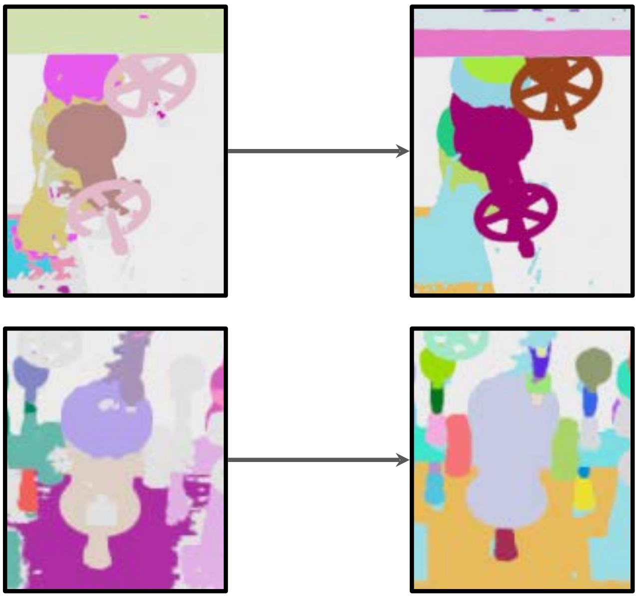








Block	purity ≥0.80
BMRF_FCC_MR_15_M_1	52%
BMRF_FCC_MR_9_M_1	47%
CHD2_MM_14	42%
CHD2_SW2_27	34%



Block	purity ≥0.80
BMRF_FCC_MR_15_M_1	52%
BMRF_FCC_MR_9_M_1	47%
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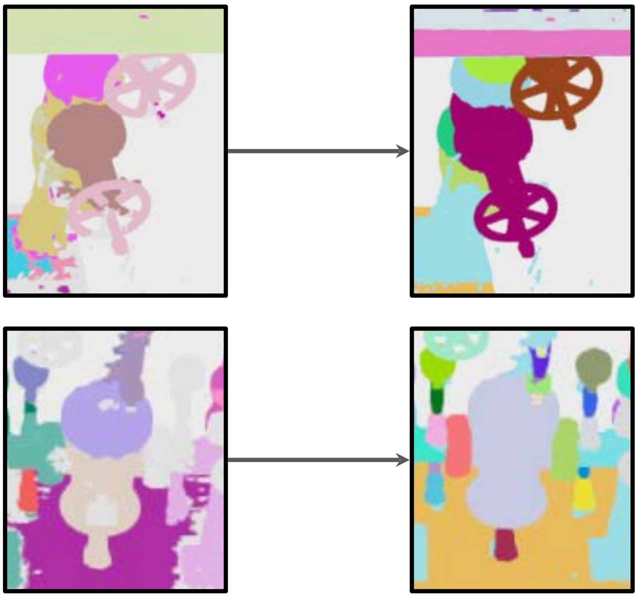


Block	purity ≥0.80
BMRF_FCC_MR_15_M_1	52%
BMRF_FCC_MR_9_M_1	47%
CHD2_MM_14	42%
CHD2_SW2_27	34%

The review

- The paper is evaluated along five axes:
  summary, strengths, weaknesses, rating, suggestions.

- Guiding principles followed:
  - weigh **novelty and impact**, not only state-of-the-art margins;
  - be **specific** about prior art;
  - keep feedback **constructive**.

Summary — The Problem

- 3D **instance** segmentation (*Mask3D*) is **closed-vocabulary**: only the ~200 trained *ScanNet200* classes.
- Can't answer ***"side table with a flower vase on it"***, affordances, materials.
- Prior **open-vocabulary** 3D work (*OpenScene*, *LERF*) is **per-point** → a heatmap, **cannot separate instances**.
- **Goal:** segment instances from arbitrary text queries , zero-shot .

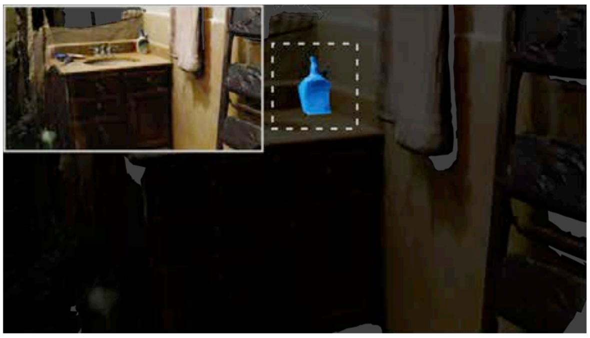
Query: “a dollhouse” *OpenScene* *OpenMask3D*

Summary

- Beats every open-vocab baseline on 6/6 metrics on *ScanNet200* — big margin on the long-tail.
- Trains nothing of its own (zero-shot feature side): *Mask3D* + *SAM* + *CLIP*.
- Build per-scene index once (5–10 min) → any text query in ~1–2 ms (cached).
- Generalizes: Out-Of-Distribution on *Replica*; *ScanNet20*-masks → *ScanNet200* novel classes.

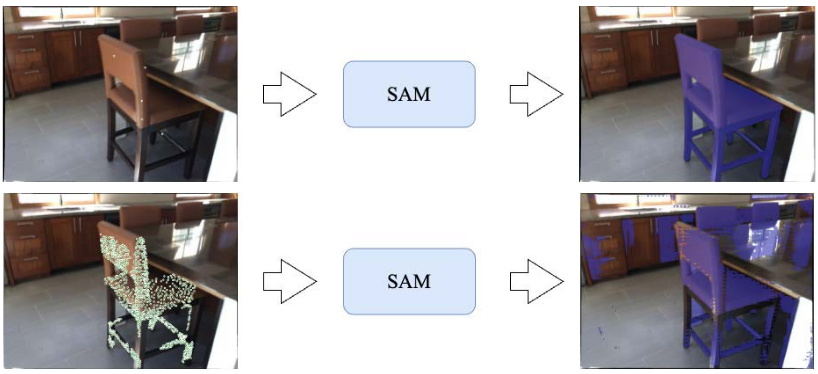
“dollhouse”

“Cetaphil”



Strengths

- Task novelty — first open-vocabulary 3D instance segmentation; beyond per-point heatmaps.
- Zero-shot / training-free features — no labels, no fine-tuning; strong on novel & long-tail (tail AP 14.9 vs ≤ 9.9).
- Clean modular design — frozen foundation models; decouples scene-encoding from query → cache once, query instantly with text or image.

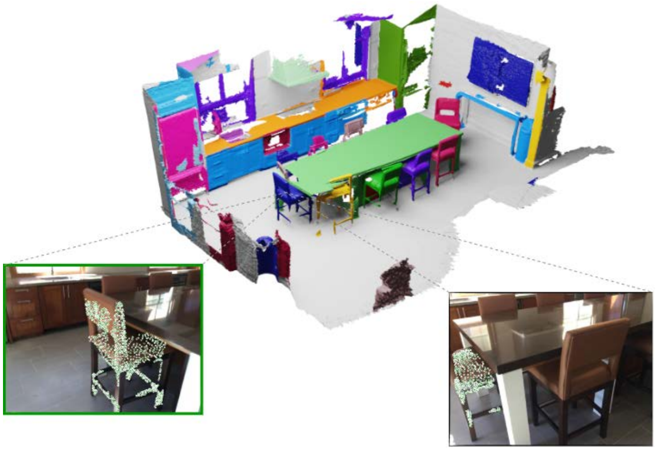


Output of SAM, using only 5 randomly sampled points (visualized as green dots) of the projected 3D mask as input.

Output of SAM, using all of the visible points from the projected 3D mask as input.

Strengths

- Generalization evidence — *Replica* + *ScanNet20*→200, not just in-distribution.
- Honest bottleneck analysis — oracle-mask study localizes the limiting factor.
- Useful ablations — *SAM* 2D masks and multi-scale crops each help.



Results

Model	Image Features	AP	AP_{50}	AP_{25}	head (AP)	common (AP)	tail (AP)
Closed-vocabulary, fully supervised							
Mask3D [58]	-	26.9	36.2	41.4	39.8	21.7	17.9
Open-vocabulary							
OpenScene [52] (2D Fusion) + masks	OpenSeg [20]	11.7	15.2	17.8	13.4	11.6	9.9
OpenScene [52] (3D Distill) + masks	OpenSeg [20]	4.8	6.2	7.2	10.6	2.6	0.7
OpenScene [52] (2D/3D Ens.) + masks	OpenSeg [20]	5.3	6.7	8.1	11.0	3.2	1.1
OpenScene [52] (2D Fusion) + masks	LSeg [43]	6.0	7.7	8.5	14.5	2.5	1.1
OpenMask3D (Ours)	CLIP [55]	**15.4**	**19.9**	**23.1**	**17.1**	**14.1**	**14.9**

Table 1: 3D instance segmentation results on the ScanNet200 validation set. Metrics are respectively: AP averaged over an overlap range, and AP evaluated at 50% and 25% overlaps. We also report AP scores for head, common, tail subsets of ScanNet200. Mask3D [58] is fully-supervised, while OpenScene [52] is built upon on 2D models (LSeg [43], OpenSeg [20]) trained on labeled datasets for 2D semantic segmentation. Since OpenScene [52] does not provide instance masks, we aggregate its per-point features using class-agnostic masks. OpenMask3D outperforms other open-vocabulary counterparts, particularly on the long-tail classes.

Results

Model	AP	AP_{50}	AP_{25}
OpenScene [52] (2D Fusion) + masks	10.9	15.6	17.3
OpenScene [52] (3D Distill) + masks	8.2	10.5	12.6
OpenScene [52] (2D/3D Ens.) + masks	8.2	10.4	13.3
OpenMask3D (Ours)	**13.1**	**18.4**	**24.2**

Table 2: 3D instance segmentation results on the Replica [61] dataset. To assess how well our model generalizes to other datasets, we use instance masks from the mask proposal module trained on ScanNet200, and test it on Replica scenes. OpenMask3D outperforms other open-vocabulary counterparts on the Replica dataset.

Model	Oracle	Img. Feat.	AP	head(AP)	common(AP)	tail(AP)
Closed-vocabulary, full sup.						
Mask3D [58]	✗	–	26.9	39.8	21.7	17.9
Mask3D [58]	✓	–	35.5	55.2	27.2	22.2
Open-vocabulary						
OpenScene [52] (2D Fusion) + masks	✓	OpenSeg [20]	22.9	26.2	22.0	20.2
OpenScene [52] (2D Fusion) + masks	✓	LSeg [43]	11.8	26.9	5.2	1.7
OpenMask3D (Ours)	✓	CLIP [55]	**29.1**	**31.1**	**24.0**	**32.9**

Table 5: 3D instance segmentation results on the ScanNet200 validation set, using oracle masks. We use ground truth instance masks for computing the per-mask features. We also report results from the fully-supervised close-vocabulary Mask3D (row 2) whose predicted masks we match to the oracle masks using Hungarian matching, and assign the predicted labels to oracle masks. On the long-tail categories, OpenMask3D outperforms even the fully supervised Mask3D with oracle masks.

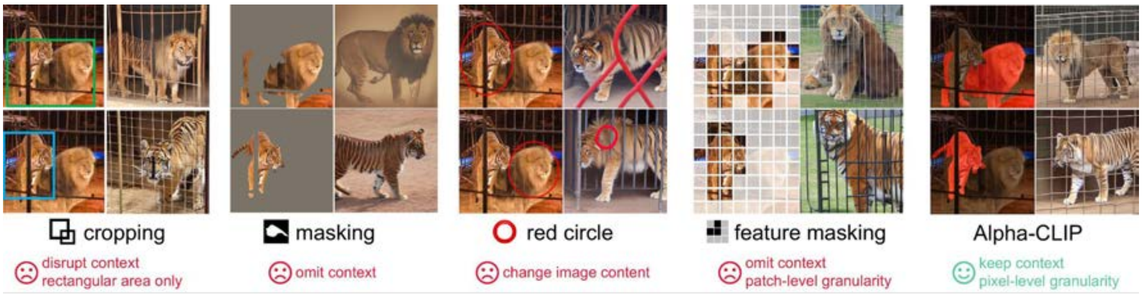
Weaknesses

- **The mask proposal is the ceiling**
  The "open-vocab" system rests on a closed-set, supervised proposer (*Mask3D*) — objects it never learned get no mask.

- **No global / relational reasoning**
  Features come only from image crops — never from inter-instance 3D relations.
  So *"the chair nearest the door"*, *"the lamp on the table"* are unanswerable.

Suggestions

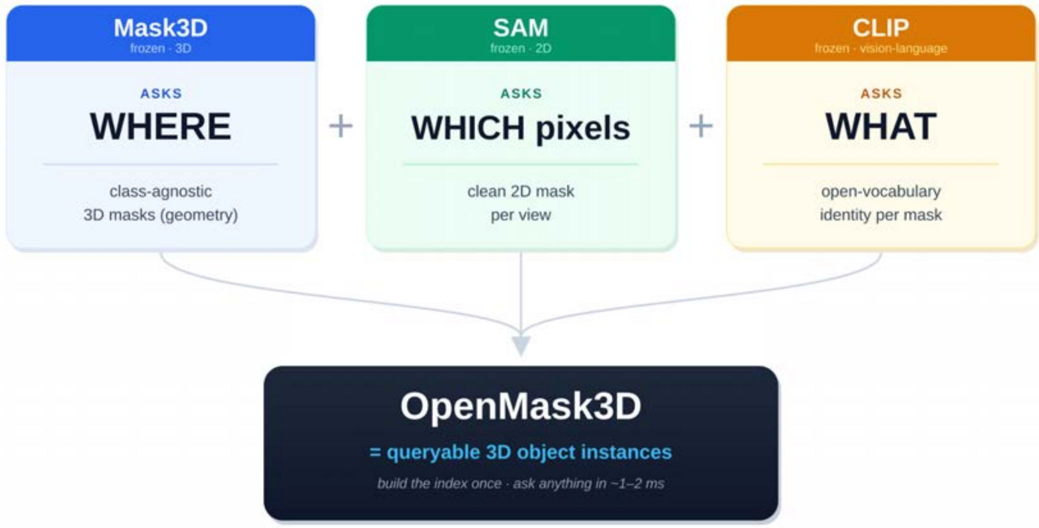
- **A class-agnostic proposer**
  Replace supervised *Mask3D* with a *SAM*-derived, label-free mask generator — *Segment3D.*
- **Reason over the 3D positions we already have**
  Enrich crops with a mask-aware CLIP (*Alpha-CLIP — CVPR2024*)



Rating & justification

Recommendation: ✅ Accept.

Innovation	9
Results	7
Writing	9
Reproducibility	7



Doesn't train a new brain — wires three pretrained ones so a closed world answers open-vocabulary questions.